



Instructor: Date:

Learner: Regional Branch:

Program: Center:

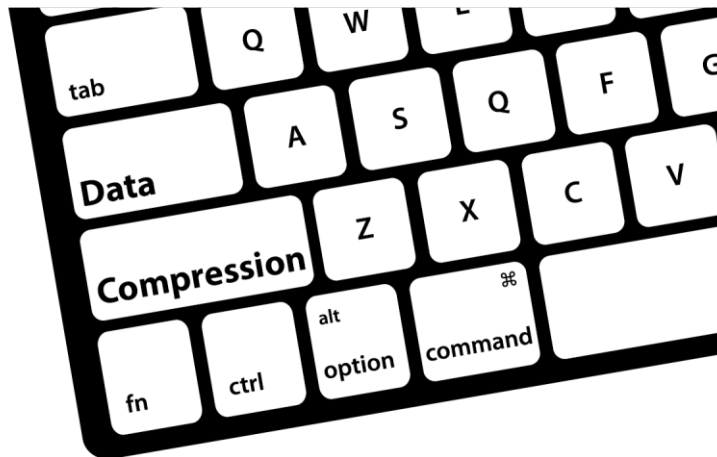
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Let's Start!

I don't know how data compression works.



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Can Do Statements:

- I can explain the concept of data compression.
- I can describe how data compression works.
- I can identify key features related to data compression.

Ask your partner:

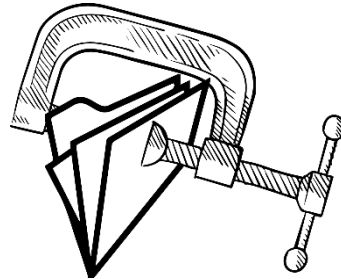
- What do you normally do when you need to attach large files to an e-mail?
- Do you understand what happens to large files when they become smaller? Explain.





Let's Explore!

Listen to Nicholas and Irene's presentation about Data Compression.



Instructions: Read the text below.

Nicholas: Hello, everybody!

Irene: Hello! To start with, Nicholas is going to introduce the concept of data compression.

Nicholas: Yes! Data compression refers to the process of removing redundant information from data to utilize less space where we store this information. Apart from saving disk space, nowadays we're compressing data to make communication faster. We normally compress heavy files to attach them to the e-mails we send, for example.

Irene: Now, I'm going to tell you a bit about how the different compression methods normally work. To easily comprehend how these methods work, it is necessary to understand a common aspect: it is not possible to find characters in equal numbers within a text. I'm thinking of an example to illustrate this idea. Let's suppose there is a text in English in front of you. If you observe the number letters e and s and the number of letters j and x there are in it, you are going to realize that j and x are less frequent. To minimize the number of bits needed to encode the text, it is necessary to assign the shortest bit codes to the most frequent characters and the longer codes to the least common characters.

Nicholas: We're going to describe some coding methods now. The Huffman coding is an algorithm that consists of a tree in which there are characters paired in ascending order starting from the least to the most common until the tree is complete.

Irene: Compared to the Huffman coding, the arithmetic coding is more efficient because it matches the probabilities of the characters to bits in a way that one same bit can represent part of more than one encoded character. Unfortunately, this method is bit slower because of the many calculations it needs.

Nicholas: Right! Another method to compress is to look for character strings that match some terms there are in a dictionary file. In this method, numbers replace matching strings to really reduce the size of most text files simply because a number is shorter than a word or phrase.

Irene: The next method is the Lempel-Ziv (LZ) compression method. In contrast to the previous one, this one doesn't use an external dictionary file. It scans the text itself to find text strings and replaces not only words but also prefixes and suffixes.

Nicholas: The last method we're going to talk about is the one that compresses graphic files. In this method, a number indicating a certain count of pixels replaces the corresponding area that represents a defined color. There are formats such as GIF which don't shrink significantly by compressing them because they are already compressed.





Let's Practice!

In a nutshell*, this is what data compression is about.

*"In a nutshell" is an idiom used to say something expresses the most relevant aspects in just a few words.

Instructions: Read the text and circle the correct words to complete the sentences.

1. Data compression refers to the process of removing redundant information from data (to utilizing / to utilize / for utilize) less space where we store this information.

2. Apart from saving disk space, we compress data (to make / make / making) communication faster.

3. We normally compress heavy files (to attached / to attach / for attached) them to the e-mails we send.



Let's Practice!

How do the different methods work?

Instructions: Read the paragraph. Circle 5 mistakes in the sentences and correct them.

Irene: After studying different methods, you is going to understand that all of them share a common aspect: it is not possible to find characters in equal numbers within a text. I going to give an example. You aren't going to do as I tell you to. Your going to observe this text in English. If you observe the number letters e and s and the number of letters j and x there are in it, you are going to realize that j and x are less frequent. To minimize the number of bits needed to encode the text, it is necessary to assign the shortest bit codes to the most frequent characters and the longer codes to the least common characters.





Extension Activity

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about the Lossy and Lossless data compression methods.
- Make a comparative table between these two concepts. Explain the purposes of using one or the other.
- Share your table with your partners.

Self-Assessment

Yes

No

1. I look for information about Lossy and Lossless data compression.

☐ ☐

2. I make a comparative table between these two concepts.

☐ ☐

3. I share the comparative table with partners.

☐ ☐


Target Words

- **redundant** (adjective): something that is repetitive
e.g. Data compression refers to the process of removing redundant information from data to utilize less space where we store this information.
- **replace** (verb): to substitute or change something for something else
e.g. In this method, numbers replace matching strings to really reduce the size of most text files simply because a number is shorter than a word or phrase.
- **shrink** (verb): to become reduced in size, amount, or value
e.g. There are formats such as GIF which don't shrink significantly by compressing them because they are already compressed.
- **algorithm** (noun): a series of steps for programming a computer to solve a problem
e.g. The Huffman coding is an algorithm that consists of a tree in which there are characters paired in ascending order starting from the least to the most common until the tree is complete.
- **character** (noun): letters, numbers, or symbols
e.g. It is not possible to find characters in equal numbers within a text.
- **reduce** (verb): to decrease the amount of something
e.g. By assigning the shortest bit codes to the most common characters the text will be reduced.



Learning Points

Infinitives of Purpose in Sentences With Simple Present and Present Continuous

To start with, it is necessary to learn to identify verbs in the infinitive form. When we say a verb is in its infinitive form, it is because it is in its basic form; that is, as you find it in a dictionary. For example: go, study, prevent, save, etc.

We can use the infinitive form of a verb to indicate a certain purpose or to answer the questions why or what for. To do so, you need to always use the preposition “**to**” before the action (verb) that expresses your purpose. There normally is another verb preceding the goal or purpose within your sentence.

Take a look at the following examples:

*There is a compression method that replaces words with numbers **to reduce** the size of files.*

*Apart from saving disk space, we compress data **to make** communication faster.*

Notice that the infinitive of purpose construction doesn’t need any comma (,), period (.), or semicolon (;).

To make a negative statement using an infinitive of purpose, we normally add **not** before **to**.

*We compress the data **not to** take up a lot of disk space.*

Examples

- *Apart from saving disk space, nowadays we’re compressing data to make communication faster.*
- *I’m thinking of an example to illustrate this idea.*
- *Data compression refers to the process of removing redundant information from data to utilize less space where we store this information.*
- *It scans the text itself to find text strings and replaces not only words but also prefixes and suffixes.*





Learning Points

Be Going to in Affirmative

We use **be going to** to talk about future plans and, in general, about events that we predict will take place in the future.

The future with **be going to** in affirmative sentences follows this structure:

Subject + verb To Be + going to + main verb (base form) + complement

Example:

*This learning object **is going to provide** you with really relevant information about data compression.*

Examples

- *Nicholas is going to introduce the concept of data compression.*
- *I'm going to tell you a bit about how the different compression methods normally work.*
- *We're going to describe some coding methods now.*
- *The last method we're going to talk about is the one that compresses graphic files.*

